



Fuhua-Parallel Set 2 · Sec 3 G3 EOY 2026 Paper 1 — Marking Scheme Worked Solutions + Marking Scheme

Cambridge Learning Group · Fuhua-parallel practice paper, Set 2 · Paper 1 ·
2026

14 pages · 80 marks

How to read this booklet:

BLACK — the question, written ON the original question paper

BLUE — the full step-by-step working and the reason for each line

RED — the marking scheme: where each B1 / M1 / A1 is earned, the traps,
and the key concept the question is testing

Mark codes: B1 independent answer mark · M1 method mark (earned even if

the answer is wrong) · A1 accuracy mark (only if the method mark was earned).

Answer all the questions.

- 1 The line $y = mx - 2$ meets the curve $y = x^2 - 3x + 7$ at two distinct points. Find the range of values of m . [4]

At the points of intersection:

$$x^2 - 3x + 7 = mx - 2$$

$$x^2 - (3+m)x + 9 = 0$$

Two distinct points, so the discriminant is positive:

$$(3+m)^2 - 4(1)(9) > 0$$

$$(m+3)^2 > 36$$

$$m+3 > 6 \text{ or } m+3 < -6$$

$$\therefore m > 3 \text{ or } m < -9$$

Key concept: TWO DISTINCT points means

$b^2 - 4ac > 0$ — not ≥ 0 , and not $b^2 - 4ac = 0$.

M1 - equates the two expressions and collects to

$$x^2 - (3+m)x + 9 = 0$$

M1 - applies $b^2 - 4ac > 0$ with

$$a = 1, b = -(3+m), c = 9$$

M1 - solves the quadratic inequality in m

$$\text{A1 - } m > 3 \text{ or } m < -9$$

Trap - writing $-9 < m < 3$ reverses the inequality — the parabola in m opens upwards, so the solution is the OUTSIDE pair of intervals.

Note - $c = 9$, not -9 : the -2 moves across as $+2$ onto the $+7$.

- (a) Express $\frac{2\sqrt{3}+5}{3-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are rational numbers. [3]

$$\frac{2\sqrt{3}+5}{3-\sqrt{3}} \times \frac{3+\sqrt{3}}{3+\sqrt{3}}$$

numerator $N = (2\sqrt{3}+5)(3+\sqrt{3})$

$$N = 6\sqrt{3} + 2(3) + 15 + 5\sqrt{3}$$

$$N = 21 + 11\sqrt{3}$$

denominator: $9 - 3 = 6$

$$\therefore \frac{21+11\sqrt{3}}{6} = \frac{7}{2} + \frac{11}{6}\sqrt{3}$$

Key concept: the conjugate of $3 - \sqrt{3}$ is $3 + \sqrt{3}$ – change the sign of the SURD term only.

M1 - multiplies by the conjugate $\frac{3+\sqrt{3}}{3+\sqrt{3}}$

M1 - expands the numerator, using $\sqrt{3} \times \sqrt{3} = 3$

A1 - $a = \frac{7}{2}, b = \frac{11}{6}$

Note - the question says **RATIONAL**, not integer – a fractional answer is expected here and must not be rounded.

- (b) Solve the equation $\sqrt{2x+3} = x-1$, giving your answer in exact form. [3]

$$\sqrt{2x+3} = x-1$$

squaring both sides: $2x+3 = x^2 - 2x+1$

$$x^2 - 4x - 2 = 0$$

$$x = \frac{4 \pm \sqrt{16+8}}{2} = 2 \pm \sqrt{6}$$

check: $x-1 \geq 0$, so $x \geq 1$

$2 - \sqrt{6} \approx -0.449$, which is rejected

$$\therefore x = 2 + \sqrt{6}$$

Key concept: squaring can create a root that does not satisfy the original equation – every answer must be checked.

M1 - squares correctly to $2x+3 = x^2 - 2x+1$

M1 - formula or completing the square on

$$x^2 - 4x - 2 = 0$$

A1 - $x = 2 \pm \sqrt{6}$

A1 - rejects $2 - \sqrt{6}$ with a reason, giving $x = 2 + \sqrt{6}$

Trap - the right-hand side is $x-1$, so the condition is $x \geq 1$ – checking only $2x+3 \geq 0$ keeps the wrong root.

- 3 It is given that $f(x) = 2x^3 + ax^2 + bx + 6$, where a and b are constants. $(x+2)$ is a factor of $f(x)$, and $f(3) = 0$.

(a) Find the value of a and of b , and hence factorise $f(x)$ completely. [4]

$$f(-2) = 0: -16 + 4a - 2b + 6 = 0$$

$$2a - b = 5 \quad \dots (1)$$

$$f(3) = 0: 54 + 9a + 3b + 6 = 0$$

$$3a + b = -20 \quad \dots (2)$$

$$(1) + (2): 5a = -15, \text{ so } a = -3$$

$$\text{from (1): } b = 2(-3) - 5 = -11$$

$$f(x) = 2x^3 - 3x^2 - 11x + 6$$

$$f(x) = (x+2)(2x^2 - 7x + 3)$$

$$\therefore f(x) = (x+2)(2x-1)(x-3)$$

Key concept: two conditions of DIFFERENT kinds — a factor and a stated value — but both give a linear equation in a and b .

M1 - uses $f(-2) = 0$ correctly

M1 - uses $f(3) = 0$ and solves the pair simultaneously

A1 - $a = -3, b = -11$

A1 - $(x+2)(2x-1)(x-3)$

Note - long division or comparing coefficients both earn the factorising mark.

(b) Hence solve the equation $2(2^{3y}) - 3(2^{2y}) - 11(2^y) + 6 = 0$, giving your answers in exact form. [3]

$$\text{let } u = 2^y, \text{ so } 2^{3y} = u^3 \text{ and } 2^{2y} = u^2$$

$$2u^3 - 3u^2 - 11u + 6 = 0, \text{ which is } f(u) = 0$$

$$(u+2)(2u-1)(u-3) = 0$$

$$u = -2 \text{ (rejected: } 2^y > 0), u = \frac{1}{2}, u = 3$$

$$2^y = \frac{1}{2} = 2^{-1}, \text{ so } y = -1$$

$$2^y = 3, \text{ so } y = \log_2 3$$

$$\therefore y = -1 \text{ or } y = \log_2 3$$

Key concept: the SAME cubic, with 2^y in place of x — spot the substitution rather than starting again.

M1 - substitutes $u = 2^y$ and recognises $f(u) = 0$

A1 - rejects $u = -2$, stating that a power of 2 is always positive

A1 - $y = -1$ and $y = \log_2 3$

Trap - giving $y = \log_2(-2)$ as a third answer — it does not exist.

Note - exact form is demanded, so $\log_2 3$ must not be written as 1.58.

(a) Show that $x^3 - x^2 + 4x - 4 = (x-1)(x^2+4)$. [2]

let $g(x) = x^3 - x^2 + 4x - 4$ and group in pairs:

$$g(x) = x^2(x-1) + 4(x-1)$$

$$g(x) = (x-1)(x^2+4) \quad (\text{shown})$$

Key concept: factorising by grouping — the common bracket $(x-1)$ appears twice.

M1 - groups as $x^2(x-1) + 4(x-1)$

A1 - reaches $(x-1)(x^2+4)$ with the working shown

Note - x^2+4 has no real factors, so it stays whole — this is the irreducible quadratic used in part (b).

(b) Hence express $\frac{2x^3+2x^2+5x+6}{x^3-x^2+4x-4}$ in partial fractions. [6]

degree 3 over degree 3, so the fraction is improper:

$$\frac{2x^3+2x^2+5x+6}{x^3-x^2+4x-4} = 2 + \frac{R(x)}{(x-1)(x^2+4)}$$

$$R(x) = (2x^3+2x^2+5x+6) - 2(x^3-x^2+4x-4)$$

$$R(x) = 4x^2 - 3x + 14$$

$$\frac{4x^2-3x+14}{(x-1)(x^2+4)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$$

$$4x^2 - 3x + 14 \equiv A(x^2+4) + (Bx+C)(x-1)$$

$$x=1: 15=5A, \text{ so } A=3$$

$$\text{compare } x^2: 4=A+B, \text{ so } B=1$$

$$\text{compare constants: } 14=4A-C, \text{ so } C=-2$$

$$\therefore 2 + \frac{3}{x-1} + \frac{x-2}{x^2+4}$$

Key concept: divide FIRST — an improper fraction has a quotient before any partial fraction is written.

M1 - identifies the fraction as improper and extracts the quotient 2

A1 - remainder $4x^2 - 3x + 14$

M1 - correct form $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$

M1 - substitution or comparison to find the constants

A1 - $A=3, B=1, C=-2$

A1 - $2 + \frac{3}{x-1} + \frac{x-2}{x^2+4}$

Trap - writing $\frac{B}{x^2+4}$ instead of $\frac{Bx+C}{x^2+4}$ loses the form mark and makes the system inconsistent.

(a) Solve the equation $\log_3(x+2) + \log_3(x-4) = 3$. [3]

$$\log_3(x+2) + \log_3(x-4) = 3$$

$$\log_3[(x+2)(x-4)] = 3$$

$$(x+2)(x-4) = 3^3 = 27$$

$$x^2 - 2x - 8 = 27$$

$$x^2 - 2x - 35 = 0$$

$$(x-7)(x+5) = 0$$

$$x = 7 \text{ or } x = -5$$

$x = -5$ makes $\log_3(x-4)$ undefined, so it is

rejected

(b) Solve the equation $4\log_9 x - \log_x 3 = 1$, giving your answers in exact form. [4]

$$\text{let } t = \log_3 x$$

$$\log_9 x = \frac{\log_3 x}{\log_3 9} = \frac{t}{2}$$

$$\log_x 3 = \frac{1}{\log_3 x} = \frac{1}{t}$$

$$4\left(\frac{t}{2}\right) - \frac{1}{t} = 1$$

$$2t - \frac{1}{t} = 1$$

$$2t^2 - t - 1 = 0$$

$$(2t+1)(t-1) = 0$$

$$t = 1 \text{ or } t = -\frac{1}{2}$$

$$\therefore x = 3 \text{ or } x = 3^{-\frac{1}{2}} = \frac{\sqrt{3}}{3}$$

Key concept: the product law first, then undo the logarithm with the base as a power.

M1 - combines to a single logarithm

M1 - writes $(x+2)(x-4) = 3^3$

A1 - $x^2 - 2x - 35 = 0$

A1 - $x = 7$ only, with $x = -5$ rejected and a reason given

Trap - keeping $x = -5$: both brackets must be

POSITIVE, so the domain is $x > 4$.

Key concept: change EVERYTHING to base 3 — two different bases cannot be combined until they match.

M1 - $\log_9 x = \frac{1}{2} \log_3 x$

M1 - $\log_x 3 = \frac{1}{\log_3 x}$ (the reciprocal law)

M1 - forms and solves $2t^2 - t - 1 = 0$

A1 - $x = 3$ and $x = \frac{\sqrt{3}}{3}$

Note - both roots are valid: $x > 0$ and $x \neq 1$ are the only restrictions.

Trap - discarding $t = -\frac{1}{2}$ as impossible — a

logarithm may be negative; it is the ARGUMENT that cannot be.

- 6 The mass, M grams, of a radioactive sample t days after it was first recorded is given by $M = M_0 e^{-kt}$, where M_0 and k are constants. The mass is 500 g when $t = 2$, and 320 g when $t = 10$.

(a) Find the value of k and of M_0 .

[4]

$$500 = M_0 e^{-2k} \quad \dots (1)$$

$$320 = M_0 e^{-10k} \quad \dots (2)$$

$$\frac{(2)}{(1)}: \frac{320}{500} = e^{-8k}$$

$$e^{-8k} = 0.64$$

$$-8k = \ln 0.64$$

$$k = 0.055786 \dots \approx 0.0558$$

$$\text{from (1): } M_0 = 500 e^{2k} = 500 e^{0.111572}$$

$$\therefore M_0 \approx 559 \text{ g}$$

(b) Find the least number of complete days, measured from $t = 0$, after which the mass first falls below 100 g. [3]

$$100 = M_0 e^{-kt}$$

$$e^{-kt} = \frac{100}{559.017}$$

$$-kt = \ln(0.178884)$$

$$t = \frac{1.720898}{0.055786} = 30.849 \dots$$

the mass is still above 100 g at $t = 30$

$\therefore$ least number of complete days: $t = 31$

Key concept: M_0 is unknown AND the readings are not at $t = 0$ — divide the two equations to eliminate it.

M1 - two correct equations from the two readings

M1 - divides to eliminate M_0 , reaching $e^{-8k} = 0.64$

A1 - $k \approx 0.0558$

A1 - $M_0 \approx 559$

Trap - assuming $M_0 = 500$ — that is the mass at $t = 2$, not at $t = 0$.

Note - keep the unrounded k in the calculator for part (b).

Key concept: a CEILING answer — the decimal is not the answer, the next whole day is.

M1 - sets $M = 100$ and takes logarithms

A1 - $t = 30.8$ (accept 30.8 to 30.9)

A1 - 31 days

Trap - rounding 30.849 to 31 by ordinary rounding is luck, not method — state that the mass first drops below 100 g DURING day 31.

Note - using the rounded $k = 0.0558$ gives 30.84, still 31 — but marks for accuracy follow through only if the method is shown.

- (a) In the expansion of $(2+kx)^6$, where k is a non-zero constant, the coefficient of x^3 is twice the coefficient of x^2 . Find the value of k . [4]

general term: $\binom{6}{r} 2^{6-r} (kx)^r$

coefficient of x^2 :

$$\binom{6}{2} 2^4 k^2 = 15(16)k^2 = 240k^2$$

coefficient of x^3 :

$$\binom{6}{3} 2^3 k^3 = 20(8)k^3 = 160k^3$$

$$160k^3 = 2(240k^2)$$

$k \neq 0$, so divide by $160k^2$: $\therefore k=3$

- (b) Hence find the coefficient of x^2 in the expansion of $(1-2x)(2+3x)^6$. [3]

from (a) the bracket is $(2+3x)^6$

coefficient of x : $\binom{6}{1} 2^5 (3) = 6(32)(3) = 576$

coefficient of x^2 :

$$\binom{6}{2} 2^4 (3^2) = 15(16)(9) = 2160$$

$(1-2x)(2+3x)^6$ gives x^2 from:

$$1 \times 2160 \text{ and } (-2x) \times 576x$$

$$1(2160) + (-2)(576) = 2160 - 1152$$

$\therefore$ coefficient of x^2 is 1008

Key concept: form BOTH coefficients in terms of k , then let the stated relation give the equation.

M1 - correct general term with both the power of 2 and the power of k

A1 - $240k^2$ and $160k^3$

M1 - forms $160k^3 = 2(240k^2)$ and divides by k^2

A1 - $k=3$

Trap - dropping the 2^{6-r} factor — the coefficients

are then $15k^2$ and $20k^3$ and k comes out as $\frac{3}{2}$.

Key concept: TWO terms contribute — the x^2 term times 1, and the x term times $-2x$.

M1 - both coefficients 576 and 2160 from the expansion

M1 - combines 1×2160 with $(-2) \times 576$

A1 - 1008

Trap - quoting 2160 alone ignores the bracket in front — one slip in (a) carries through both marks here.

- 8 The points $A(-2, 3)$ and $B(6, 7)$ are joined by a straight line. Solutions by accurate drawing will not be accepted.

(a) Find the equation of the perpendicular bisector of AB . [3]

midpoint of AB : $\left(\frac{-2+6}{2}, \frac{3+7}{2}\right) = (2, 5)$

gradient of AB : $\frac{7-3}{6-(-2)} = \frac{4}{8} = \frac{1}{2}$

perpendicular gradient: -2

$y - 5 = -2(x - 2)$

$\therefore y = -2x + 9$

Key concept: a perpendicular bisector needs BOTH the midpoint and the negative reciprocal gradient.

B1 - midpoint $(2, 5)$

M1 - gradient $\frac{1}{2}$, then perpendicular gradient -2

A1 - $y = -2x + 9$

Note - $m_1 m_2 = -1$ is the test; $\frac{1}{2} \times (-2) = -1$ ✓

(b) The perpendicular bisector of AB cuts the x -axis at the point C . Find the coordinates of C . [2]

on the x -axis, $y = 0$:

$0 = -2x + 9$

$x = \frac{9}{2}$

$\therefore C\left(\frac{9}{2}, 0\right)$

Key concept: the x -axis is the line $y = 0$.

M1 - substitutes $y = 0$ into the bisector

A1 - $C\left(\frac{9}{2}, 0\right)$

Note - leave it as $\frac{9}{2}$ — 4.5 is accepted, 4 is not.

(c) Explain why triangle ABC is isosceles.

[2]

C lies on the perpendicular bisector of AB .
Every point on the perpendicular bisector of AB is equidistant from A and B .
 $\therefore CA=CB$, so triangle ABC is isosceles.

Key concept: this is the REASONING mark – quote the equidistance property, do not merely compute.

B1 - states that C lies on the perpendicular bisector of AB

B1 - concludes $CA=CB$, hence isosceles

Note - a full distance calculation showing $CA=CB$ scores both marks, but the one-line property argument is what the examiner wants.

(d) The point D lies on the perpendicular bisector of AB and the area of triangle ABD is 30 units². Find the possible coordinates of D .

[5]

$$AB = \sqrt{8^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$$

$$\text{area} = \frac{1}{2} \times AB \times h = 30$$

$$h = \frac{60}{4\sqrt{5}} = \frac{15}{\sqrt{5}} = 3\sqrt{5}$$

D is on the bisector, so h is measured from the midpoint $M(2,5)$

along the bisector, whose direction is $\frac{(1, -2)}{\sqrt{5}}$

$$D = (2, 5) \pm 3\sqrt{5} \times \frac{(1, -2)}{\sqrt{5}} = (2, 5) \pm 3(1, -2)$$

$$\therefore D(5, -1) \text{ or } D(-1, 11)$$

Key concept: the perpendicular bisector is already perpendicular to AB , so the distance from D to

AB is just MD .

M1 - $AB = 4\sqrt{5}$

M1 - uses $\text{area} = \frac{1}{2} \times AB \times h$ to get $h = 3\sqrt{5}$

M1 - measures $3\sqrt{5}$ from $M(2,5)$ along the bisector

A1 - $D(5, -1)$

A1 - $D(-1, 11)$

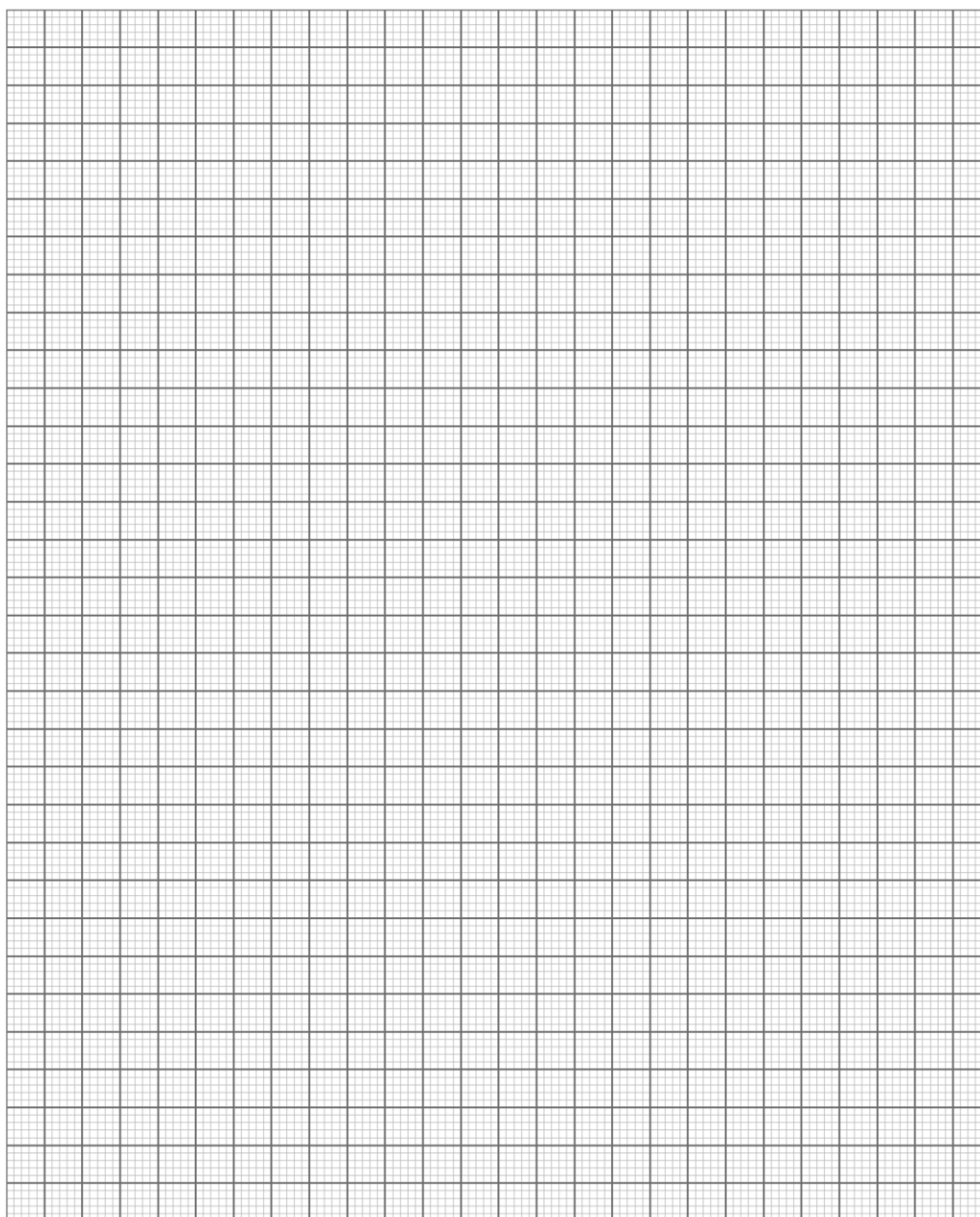
Trap - giving only one point – **POSSIBLE** coordinates means both sides of AB must be given.

Note - the shoelace formula with $D(x, -2x+9)$ also works and earns the same marks.

- 9 The variables x and y are related by the equation $y = ax^b$, where a and b are constants. The table shows measured values of x and y .

x	1	4	16	36	100
y	3.00	6.00	12.0	18.0	30.0

- (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



$$y = ax^b$$

$$\lg y = \lg a + b \lg x$$

plot $\lg y$ against $\lg x$: gradient b , vertical intercept $\lg a$.

$$\lg x: 0, 0.602, 1.204, 1.556, 2.000$$

$$\lg y: 0.477, 0.778, 1.079, 1.255, 1.477$$

Key concept: take logarithms of BOTH variables — a power law needs $\lg x$, not x .

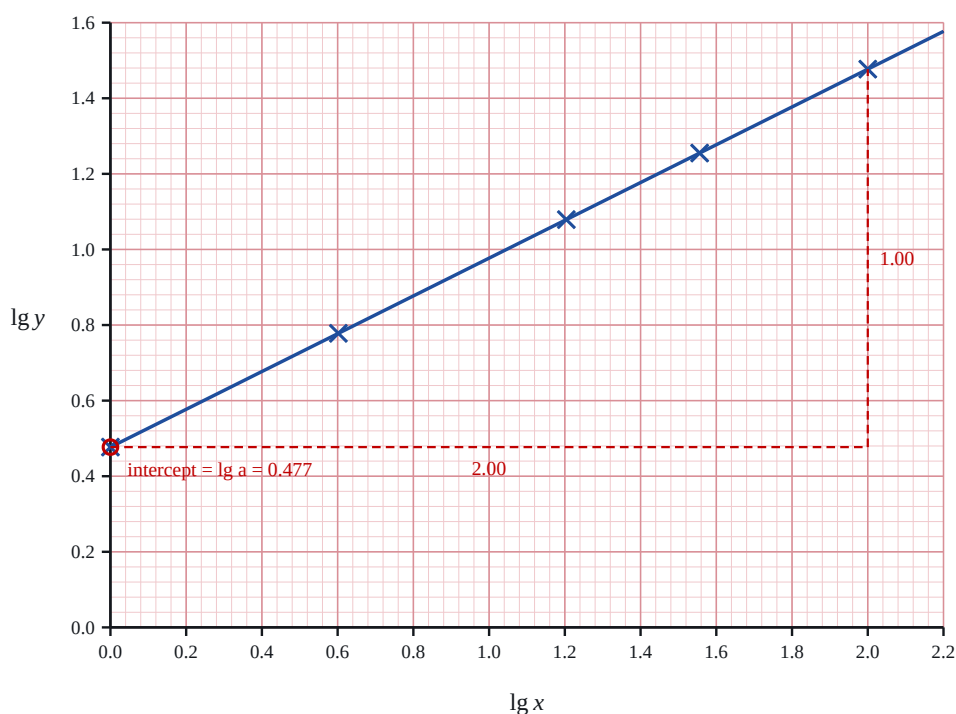
$$\text{B1} - \lg y = \lg a + b \lg x$$

B1 - states: plot $\lg y$ against $\lg x$, gradient b , intercept $\lg a$

B1 - both logarithm rows computed and a ruled straight line drawn

Trap - plotting $\lg y$ against x suits $y = ab^x$, not $y = ax^b$.

- 9 The variables x and y are related by $y = ax^b$, where a and b are constants. (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



Gradient $b = 1.00 \div 2.00 = 0.5$ · vertical intercept $\lg a = 0.477$, so $a = 10^{0.477} = 3$ · the ruled line is $\lg y = 0.477 + 0.5 \lg x$

WORKING — BLUE INK

$$y = ax^b \Rightarrow \lg y = \lg a + b \lg x$$

Comparing with $Y = mX + c$: plot $\lg y$ against $\lg x$;
gradient b , vertical intercept $\lg a$.

$\lg x$	0	0.602	1.204	1.556	2.000
$\lg y$	0.477	0.778	1.079	1.255	1.477

MARKS & NOTES — RED INK

Key concept: take logarithms of *both* variables — a power law needs $\lg x$, not x .

- **B1** $\lg y = \lg a + b \lg x$
- **B1** states: plot $\lg y$ against $\lg x$, gradient b , intercept $\lg a$
- **B1** both logarithm rows computed to 3 d.p., all five points plotted and a ruled straight line drawn
- **Trap** plotting $\lg y$ against x suits $y = ab^x$, not $y = ax^b$
- **Note** the intercept is $\lg a$, not a — part (b) still needs $a = 10^{0.477}$
- **Note** the points are exactly collinear (the data comes from $y = 3\sqrt{x}$), so any visible scatter means a logarithm was mis-read

(b) Use your graph to estimate the value of a and of b .

[3]

$$\text{gradient } b = \frac{1.477 - 0.477}{2.000 - 0} = \frac{1.000}{2.000}$$

$$b \approx 0.5$$

$$\text{vertical intercept} = \lg a = 0.477$$

$$a = 10^{0.477}$$

$$\therefore a \approx 3, b \approx 0.5$$

Key concept: the intercept is $\lg a$, NOT a — one more step is needed.

M1 - gradient from two points on the drawn line

A1 - $b \approx 0.5$

M1 - reads the intercept and applies $a = 10^{\text{intercept}}$

A1 - $a \approx 3$

Note - accept a from 2.8 to 3.2 and b from 0.47 to 0.53.

Note - check: $y = 3\sqrt{x}$ gives $y = 12$ at $x = 16$ ✓

- 10 Given that $\sin C = q$, where C is reflex and $180^\circ < C < 270^\circ$, express each of the following in terms of q .

(i) $\cos C$

[2]

$180^\circ < C < 270^\circ$, so C is in quadrant 3 in quadrant 3 both sine and cosine are negative

$$\sin^2 C + \cos^2 C = 1$$

$$\cos^2 C = 1 - q^2$$

$$\therefore \cos C = -\sqrt{1 - q^2}$$

Key concept: the quadrant fixes the SIGN; the identity fixes only the size.

M1 - uses $\sin^2 C + \cos^2 C = 1$

A1 - $\cos C = -\sqrt{1 - q^2}$, negative sign justified by quadrant 3

Note - q is itself negative here, since $\sin C < 0$ in quadrant 3.

(ii) $\tan C$

[2]

$$\tan C = \frac{\sin C}{\cos C}$$

$$\tan C = \frac{q}{-\sqrt{1 - q^2}}$$

$$\therefore \tan C = -\frac{q}{\sqrt{1 - q^2}}$$

Key concept: build tan from the two ratios already found — no new triangle.

M1 - $\tan C = \frac{\sin C}{\cos C}$ with both substituted

A1 - $-\frac{q}{\sqrt{1 - q^2}}$

Note - this is **POSITIVE**, because $q < 0$ — tangent is positive in quadrant 3, a useful check.

(iii) $\sec(360^\circ - C)$

[2]

$$\cos(360^\circ - C) = \cos C$$

$$\sec(360^\circ - C) = \frac{1}{\cos C}$$

$$\sec(360^\circ - C) = \frac{1}{-\sqrt{1 - q^2}}$$

$$\therefore \sec(360^\circ - C) = -\frac{1}{\sqrt{1 - q^2}}$$

Key concept: $360^\circ - C$ is a quadrant-4 angle, where cosine keeps its sign — so the value is simply $\cos C$.

M1 - $\cos(360^\circ - C) = \cos C$

A1 - $-\frac{1}{\sqrt{1 - q^2}}$

Trap - writing $\sec(360^\circ - C) = \sec C$ and then leaving it there — the answer must be in terms of q .

(a) Prove the identity $\frac{1-\cos\theta}{\sin\theta} + \frac{\sin\theta}{1-\cos\theta} = 2\operatorname{cosec}\theta$. [3]

$$\text{LHS} = \frac{(1-\cos\theta)^2 + \sin^2\theta}{\sin\theta(1-\cos\theta)}$$

using $\cos^2\theta + \sin^2\theta = 1$:

$$\text{LHS} = \frac{2-2\cos\theta}{\sin\theta(1-\cos\theta)} = \frac{2(1-\cos\theta)}{\sin\theta(1-\cos\theta)}$$

$$\text{LHS} = \frac{2}{\sin\theta} = 2\operatorname{cosec}\theta = \text{RHS (proved)}$$

Key concept: work on ONE side only, and let $\sin^2\theta + \cos^2\theta = 1$ collapse the numerator.

M1 - single fraction over $\sin\theta(1-\cos\theta)$

M1 - applies $\cos^2\theta + \sin^2\theta = 1$ to reach $2-2\cos\theta$

A1 - cancels $(1-\cos\theta)$ and concludes $2\operatorname{cosec}\theta$

Trap - cross-multiplying both sides is not a proof – it assumes the result.

(b) Hence solve the equation $\frac{1-\cos\theta}{\sin\theta} + \frac{\sin\theta}{1-\cos\theta} = 4$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

by part (a): $2\operatorname{cosec}\theta = 4$

$$\operatorname{cosec}\theta = 2$$

$$\sin\theta = \frac{1}{2}$$

basic angle = 30° ; sine is positive in quadrants 1 and 2

$$\theta = 30^\circ \text{ or } \theta = 180^\circ - 30^\circ$$

$$\therefore \theta = 30^\circ \text{ or } \theta = 150^\circ$$

Key concept: HENCE – use the identity, do not restart from the fraction.

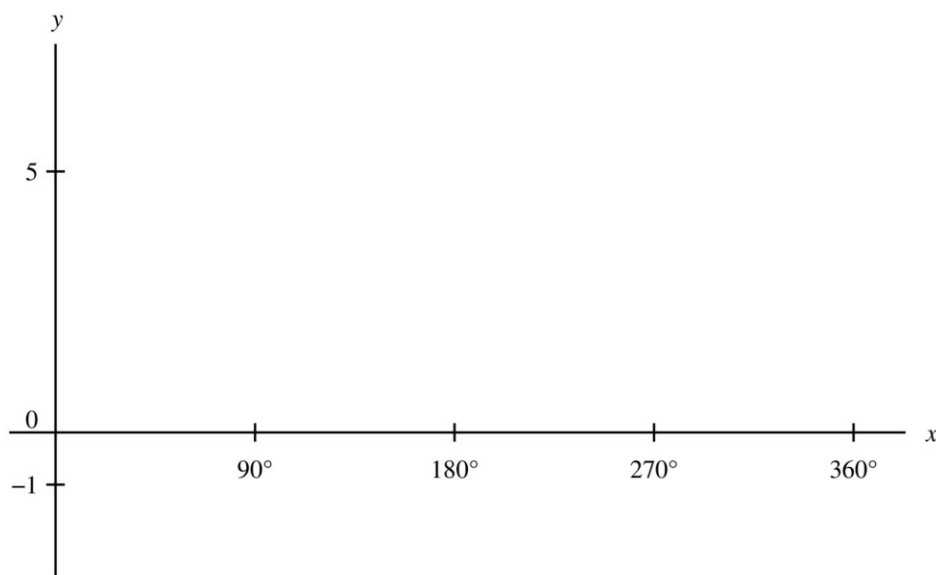
M1 - cosec $\theta = 2$, hence $\sin\theta = \frac{1}{2}$

A1 - $\theta = 30^\circ$

A1 - $\theta = 150^\circ$

Note - $\theta = 0^\circ$ and $\theta = 180^\circ$ are excluded anyway – the original expression is undefined there.

- (c) Sketch the graph of $y=2+3\sin 2x$ for $0^\circ \leq x \leq 360^\circ$, stating the amplitude and the period of the curve. [4]



----- End of Paper -----

$$y = 2 + 3 \sin 2x$$

amplitude = 3

$$\text{period} = \frac{360^\circ}{2} = 180^\circ$$

$$\text{maximum} = 2 + 3 = 5$$

$$\text{minimum} = 2 - 3 = -1$$

the curve starts at $(0, 2)$, rising

two complete cycles on $0^\circ \leq x \leq 360^\circ$

maxima at $x = 45^\circ$ and $x = 225^\circ$

minima at $x = 135^\circ$ and $x = 315^\circ$

Key concept: $\sin 2x$ halves the period, and the +2 lifts the whole curve — the amplitude is unchanged.

B1 - amplitude 3

B1 - period 180°

B1 - two complete cycles drawn, starting and ending at $y = 2$

B1 - maximum 5 and minimum -1 correctly placed

Trap - drawing one cycle — the period is 180° , so 360° holds exactly two.